

Sri Lanka Institute of Information Technology



Lab Submission
02

IT24610814

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Mathematics for Computing | IT1130

B.Sc. (Hons) in Information Technology

Exercise 01

03)

1.

```
>> x = 28;  
>> y = 42;  
>> result = gcd(x, y);  
>> disp(result);
```

14

2.

```
>> x = 36;  
>> y = 60;  
>> result = gcd(x, y);  
>> disp(result);
```

12

3.

```
>> x = 81;  
>> y = 153;  
>> result = gcd(x, y);  
>> disp(result);
```

9

4.

```
>> x = 48;  
>> y = 72;  
>> z = 120;  
>> result = gcd(x,y,z);  
>> disp(result);
```

24

5.

```
>> x = 121;  
>> y = 88;  
>> result = gcd(x,y);  
>> disp(result);  
11
```

04)

1.

```
>> x = 12;  
>> y = 15;  
>> z = 25;  
>> result = lcm(x,y,z);  
>> disp(result);  
300
```

2.

```
>> x = 20;  
>> y = 30;  
>> result = lcm(x,y);  
>> disp(result);  
60
```

3.

```
>> x = 9;  
>> y = 14;  
>> result = lcm(x,y);  
>> disp(result);  
126
```

4.

```
>> x = 16;  
>> y = 24;  
>> z = 40;  
>> result = lcm(x,y,z);  
>> disp(result);  
240
```

5.

```
>> x = 35;  
>> y = 50;  
>> result = lcm(x,y);  
>> disp(result);  
350
```

05)

1.

```
>> a = 12;  
>> b = 5;  
>> result = mod(a, b);  
>> disp(result);  
2
```

2.

```
>> a = 20;  
>> b = 7;  
>> result = mod(a, b);  
>> disp(result);  
6
```

3.

```
>> a = 9;  
>> b = 3;  
>> result = mod(a, b);  
>> disp(result);  
0
```

4.

```
>> a = 16;  
>> b = 9;  
>> result = mod(a, b);  
>> disp(result);  
7
```

5.

```
>> a = 35;  
>> b = 11;  
>> result = mod(a, b);  
>> disp(result);  
2
```

06)

1.

```
>> a = 12;  
>> b = 2;  
>> result = a^b;  
>> disp(result);  
144
```

2.

```
>> a = 2;  
>> b = 7;  
>> result = a^b;  
>> disp(result);  
128
```

3.

```
>> a = 9;  
>> b = 3;  
>> result = a^b;  
>> disp(result);  
729
```

4.

```
>> a = 2;  
>> b = 2;  
>> c = 2;  
>> d = 8;  
>> result = (a^b)*(c^d);  
>> disp(result);  
1024
```

5.

```
>> a = 3;  
>> b = 8;  
>> c = 2;  
>> result = (a^b)^c;  
>> disp(result);  
4.3047e+07(43046721)
```

07)

```
>> num = 3/5 + 7/9;
```

```
>> disp(num);
```

```
1.3778
```

Exercise 02

01)

1.

```
>> fraction1 = 5/4;
```

```
>> fraction2 = 1/2;
```

```
>> fraction3 = 5/6;
```

```
>> sum_fractions = fraction1 + fraction2 + fraction3;
```

```
>> [num, den] = rat(sum_fractions);
```

```
>> fprintf('%d/%d\n', num, den);
```

```
31/12
```

2.

```
fraction1 = 5/6;
```

```
fraction2 = 3/2;
```

```
sum_fractions = fraction1 + fraction2;
```

```
[num, den] = rat(sum_fractions);
```

```
fprintf('%d/%d\n', num, den);
```

```
7/3
```

3.

4.

5.

02)

1.

```
>> a = 5/4;  
>> b = 1/2;  
>> result = a*b;  
>> disp(result);  
0.6250
```

2.

3.

```
>> a = 6/5;  
>> b = 1/7;  
>> result = a*b;  
>> disp(result);  
0.1714
```

03)

1.

```
>> a = 5/4;  
>> b = 1/2;  
>> result = a/b;  
>> disp(result);  
2.5000
```

2.

3.

```
>> a = 6/5;  
>> b = 1/7;  
>> result = a/b;  
>> disp(result);  
8.4000
```